

Lecture 07

Health Informatics

Microsoft Excel



Dr. Nabeel Abdulrazzaq Fattah

Assistant Professor Biomedical Engineering, PhD, UK

Nursing and patient care, Diploma, Ireland

Nursing Studies - The Physical Examination, Diploma, Ireland



Advanced Excel

Content

- **Advanced Formulas:**

- Nested functions, logical functions (IF, AND, OR)
- Lookup functions (VLOOKUP, HLOOKUP)

- **Data Validation:**

- Creating data entry rules

- **Macros:**

- Recording and running macros to automate tasks

Advanced Formulas

- Nested Functions: Combine multiple functions within a formula.
- Logical Functions: IF, AND, OR, NOT.
- Lookup Functions: VLOOKUP, HLOOKUP, INDEX, MATCH.

Nested Functions Example

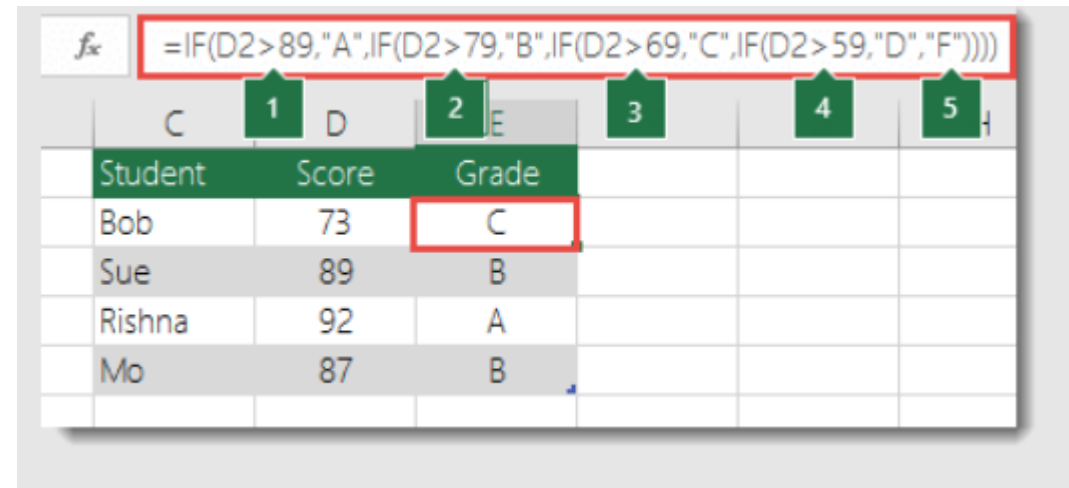
- Example: Calculate the commission based on sales and a commission rate.
- Formula: `=IF(A2>10000, A2*0.1, A2*0.05)`

- * “Nesting” refers to the practice of joining multiple functions together in one formula.

=IF(D2>89,"A",IF(D2>79,"B",IF(D2>69,"C",IF(D2>59,"D","F"))))

This complex nested IF statement follows a straightforward logic:

1. If the Test Score (in cell D2) is greater than 89, then the student gets an A
2. If the Test Score is greater than 79, then the student gets a B
3. If the Test Score is greater than 69, then the student gets a C
4. If the Test Score is greater than 59, then the student gets a D
5. Otherwise the student gets an F



The screenshot shows an Excel spreadsheet with a formula bar at the top containing the nested IF formula: `=IF(D2>89,"A",IF(D2>79,"B",IF(D2>69,"C",IF(D2>59,"D","F"))))`. Below the formula bar is a table with columns labeled C, D, E, 3, 4, and 5. The first row of the table has headers: Student, Score, Grade. The second row shows Bob with a score of 73 and a grade of C. The third row shows Sue with a score of 89 and a grade of B. The fourth row shows Rishna with a score of 92 and a grade of A. The fifth row shows Mo with a score of 87 and a grade of B. The cell containing the grade 'C' for Bob is highlighted with a red border.

Student	Score	Grade
Bob	73	C
Sue	89	B
Rishna	92	A
Mo	87	B

Logical Functions Example

- Example: Determine if a student passed or failed based on their grade.
- Formula: =IF(B2>=60, "Pass", "Fail")

	A	B	C	D
1	Name	Marks	Result	
2	M Ameer	70	Pass	
3	K Samad	59	Fail	
4	L Kamaran	85	Pass	
5				

VLOOKUP Function

- Purpose: Look up a value in a table.
- Syntax: VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup])

VLOOKUP Example

	A	B	C	D	E
1	ID	Last name	First name	Title	Birth date
2	101	Davis	Sara	Sales Rep	12/08/68
3	102	Fontana	Olivier	VP (Sales)	02/19/52
4	103	Leal	Karina	Sales Rep	08/30/63
5	104	Patten	Michael	Sales Rep	09/19/58
6	105	Burke	Brian	Sales Manager	03/04/55
7	106	Sousa	Luis	Sales Rep	07/02/63
8					
9					
10	Formula	=VLOOKUP(B3,B2:E7,2,FALSE)			
11	Result	Olivier			
12					

VLOOKUP looks for *Fontana* in the first column (column B) in table_array B2:E7, and returns *Olivier* from the second column (column C) of the table_array. FALSE returns an exact match.

HLOOKUP Function

- Purpose: Look up a value in a row.
- Syntax: HLOOKUP(lookup_value, table_array, row_index_num, [range_lookup])

HLOOKUP Example

Looks up "Axles" in row 1, and returns the value from row 2 that's in the same column (column A).

E2

✖

✓

fx

=HLOOKUP("Axles", A1:C4, 2, TRUE)

	A	B	C	D	E	F	G
1	Axles	Bearings	Bolts		Hlookup		
2	4	4	9		4		
3	5	7	10				
4	6	8	11				
5							
6							
7							

INDEX and MATCH Functions

- Purpose: Combine to look up a value in a table.
- Syntax:

INDEX(array, row_num, [column_num]) and
MATCH(lookup_value, lookup_array, [match_type])

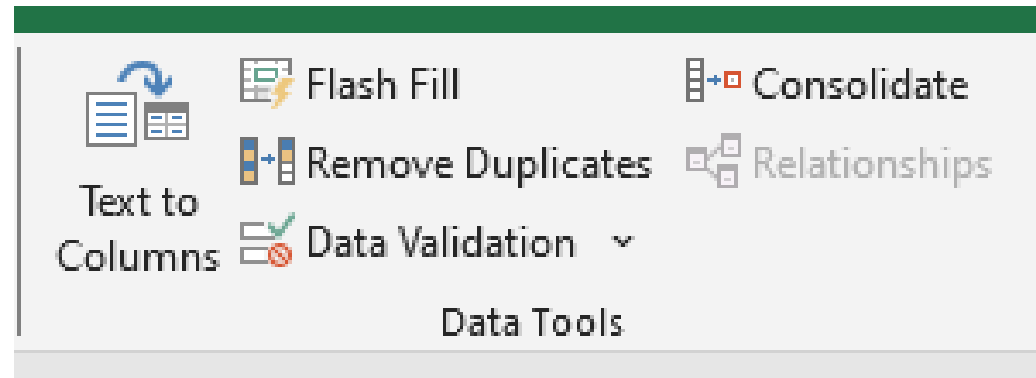
INDEX and MATCH Example

The intersection of the second row and third column in the range A2:C6, which is the contents of cell C3.

E2						
=INDEX(A2:C6, 2, 3)						
	A	B	C	D	E	F
1	Fruit	Price	Count		Index	
2	Apples	\$0.69	40		38	
3	Bananas	\$0.34	38			
4	Lemons	\$0.55	15			
5	Oranges	\$0.25	25			
6	Pears	\$0.59	40			
7						
8	Almonds	\$2.80	10			
9	Cashews	\$3.55	16			
10	Peanuts	\$1.25	20			
11	Walnuts	\$1.75	12			

Data Validation

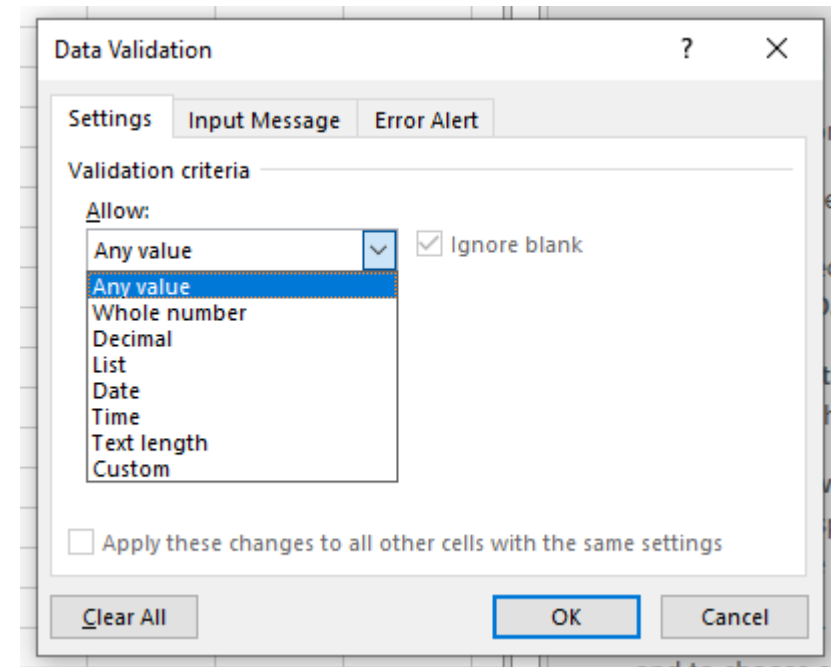
- Purpose: Restrict data entry to specific values or criteria.
- Types:
 - Data validation lists
 - Decimal places
 - Whole numbers
 - Date ranges



Applying Data Validation

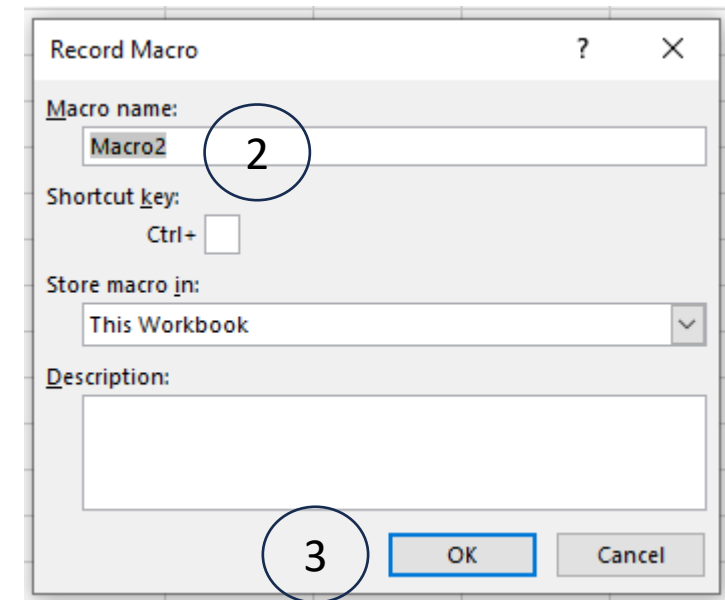
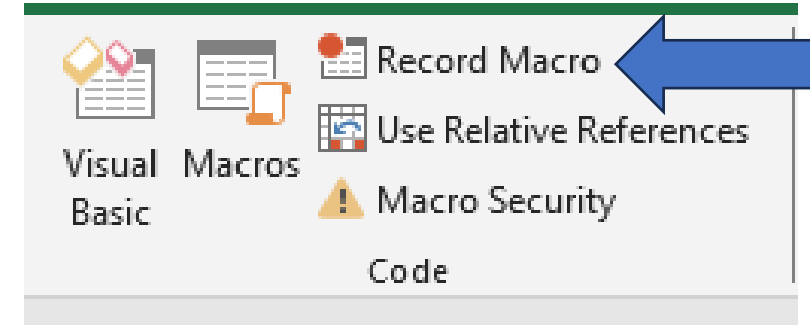
Steps:

1. Select the cell range.
2. Go to the "Data" tab.
3. Click "Data Validation."
4. Set the validation criteria.



Macros

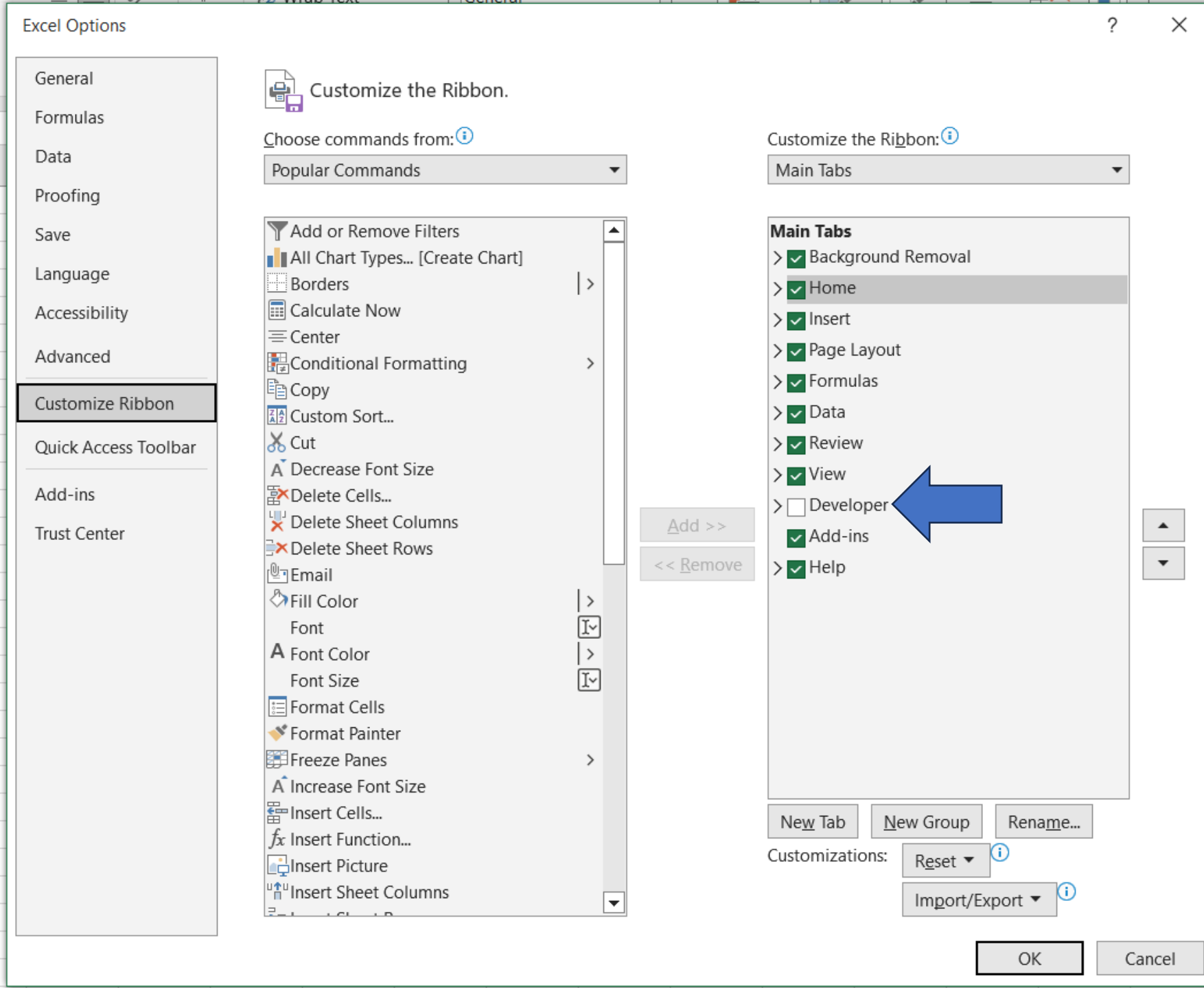
- Purpose: Automate repetitive tasks.
- Recording Macros:
 - Turn on macro recording.
 - Perform the desired actions.
 - Stop macro recording.



Running Macros

Steps:

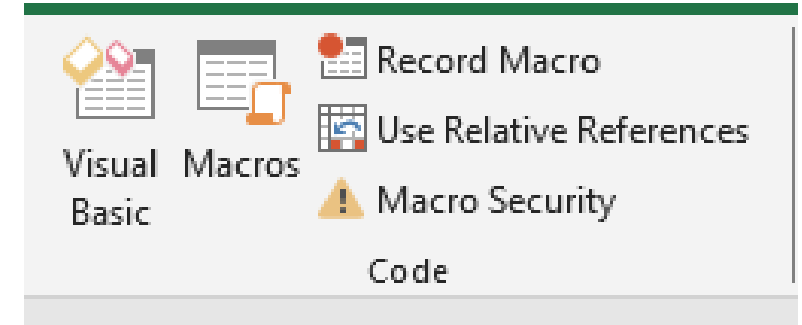
1. Go to the "Developer" tab.
2. Click "Macros."
3. Choose the macro and run it.



Editing Macros

Steps:

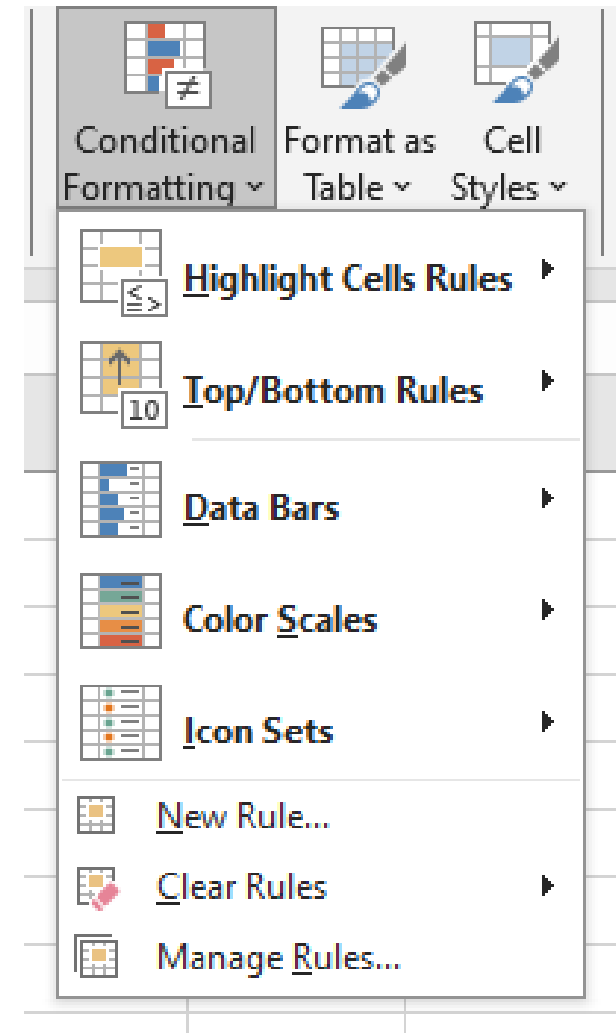
1. Go to the "Developer" tab.
2. Click "Visual Basic."
3. Edit the macro code.



Conditional Formatting

Advanced Techniques:

- Highlight cells rules
- Data bars
- Color scales
- Icon sets



Pivot Tables

Advanced Techniques:

- Calculated fields
- Slicers
- Timelines

A	B	C	D	E	F	G	H	I	J	K	L
#	Name	date	Salary								
1	Ahmed J	01/02/2012	522								
2	Muhammed Q	01/05/2022	122								
3	Rand K	01/06/2024	156								
4	Safa A	01/02/2014	544								

date	(All)
Row Labels	Sum of Salary
fdg fg	156
jpgj fg f fg	122
jdfjfdv gj skgjs	522
sfdg f	544
Grand Total	1344

PivotTable Fields

Choose fields to add to report:

Search

- ☐ #
- ☒ Name
- ☒ date
- ☒ Salary
- ☐ Months (date)
- ☐ Quarters (date)

Drag fields between areas below:

Filters

date

Columns

Rows

Name

Values

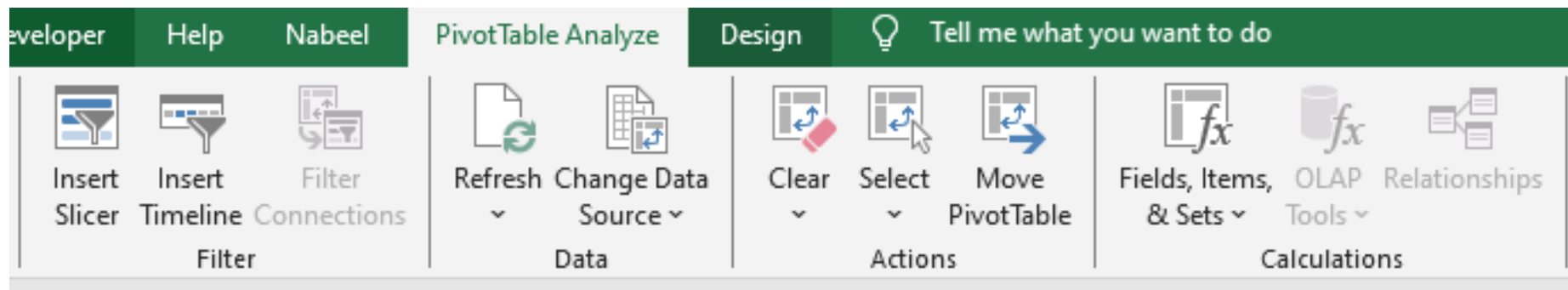
Sum of Salary

☐ Defer Layout Update Update

Creating Calculated Fields

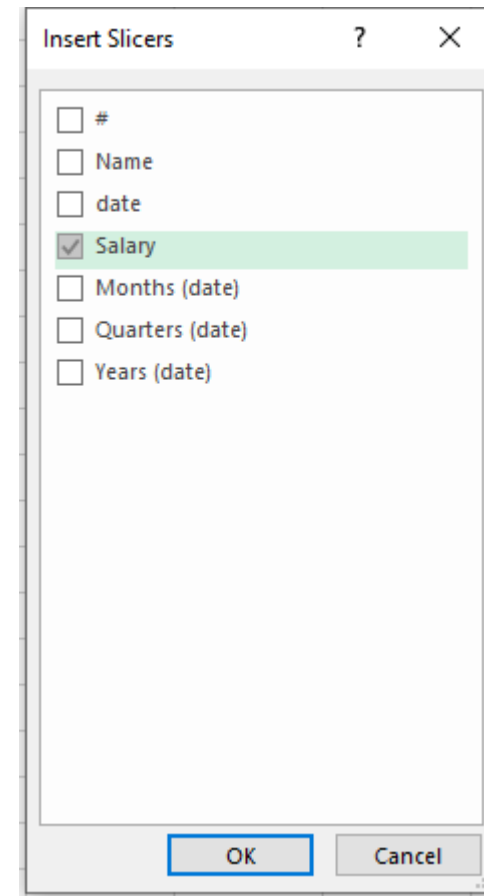
Steps:

1. Go to the "Analyze" tab.
2. Click "Fields, Items & Sets."
3. Choose "Calculated Field."
4. Create a formula.



Using Slicers

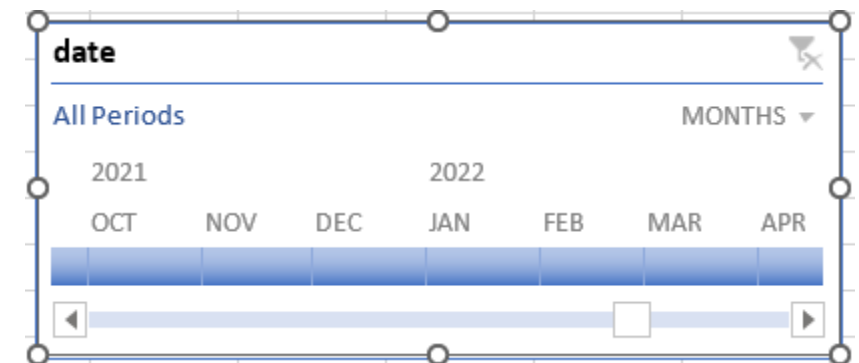
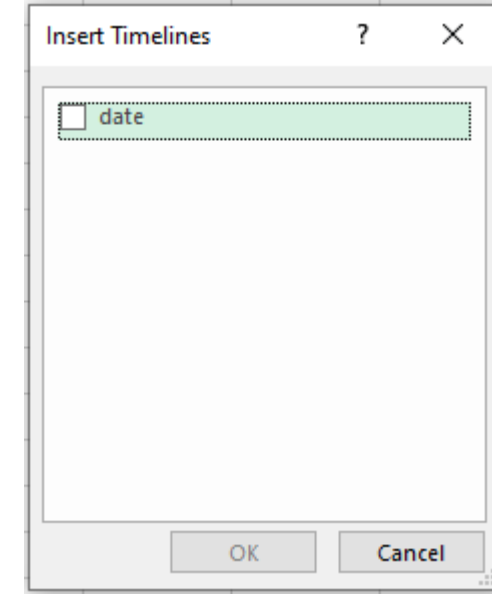
- Purpose: Filter pivot table data interactively.
- Steps:
 1. Go to the "Analyze" tab.
 2. Click "Insert Slicer."
 3. Choose the field(s) to slice.



Salary
122
156
522
544

Using Timelines

- Purpose: Filter pivot table data by date or time.
- Steps:
 1. Go to the "Analyze" tab.
 2. Click "Insert Timeline."
 3. Choose the date or time field.



Exercise 1: Creating a Complex Formula

Task: Calculate the total cost based on price, quantity, and tax.

Instructions:

- Use nested functions and cell references.

Exercise 2: Applying Data Validation

Task: Restrict data entry in a cell to a specific list of values.

Instructions:

- Apply data validation and create a list.

Exercise 3: Creating a Macro

Task: Automate the process of calculating a sum.

Instructions:

- Record a macro and edit the code if necessary.

Thanks

